

Section 5.4 : [#1.\(a\)~\(g\)](#), [#2.\(c\),\(e\)](#), [#6.\(a\),\(d\)](#), [#9.\(a\),\(d\)](#), [#10.\(a\),\(d\)](#), [#17](#), [#18](#)

*Click the problem number above.

1. Label the following statements as true or false.

(a) There exists a linear operator T with no T -invariant subspace.

Solution: False

Any linear operator T has a trivial T -invariant subspace $\{0\}$.

(b) If T is a linear operator on a finite-dimensional vector space V and W is a T -invariant subspace of V , then the characteristic polynomial of T_W divides the characteristic polynomial of T .

Solution: True

See Theorem 5.21.

(c) Let T be a linear operator on a finite-dimensional vector space V , and let v and w be in V . If W is the T -cyclic subspace generated by v , W' is the T -cyclic subspace generated by w , and $W = W'$, then $v = w$.

Solution: False

Let us give a counterexample. Let $V = \mathbb{R}^2$ and $T((a, b)) = (b, a)$. If $v = (1, 0)$ and $w = (0, 1)$, then $W = W' = V$ but $v \neq w$.

(d) If T is a linear operator on a finite-dimensional vector space V , then for any $v \in V$ the T -cyclic subspace generated by v is the same as the T -cyclic subspace generated by $T(v)$.

Solution: False

Let us give a counterexample. Let $V = \mathbb{R}^2$ and $T((a, b)) = (0, b)$. If $v = (1, 0)$, then we have $T(v) = (0, 0)$ so the T -cyclic subspace generated by v is $\langle v \rangle$ but the T -cyclic subspace generated by $T(v)$ is $\{0\}$.

- (e) Let T be a linear operator on an n -dimensional vector space. Then there exists a polynomial $g(t)$ of degree n such that $g(T) = T_0$.

Solution: True

Let $g(t)$ be the characteristic polynomial of T by Theorem 5.23.

- (f) Any polynomial of degree n with leading coefficient $(-1)^n$ is the characteristic polynomial of some linear operator.

Solution: True

See Exercise 19. The linear operator we are looking for is L_A .

- (g) If T is a linear operator on a finite-dimensional vector space V , and if V is the direct sum of k T -invariant subspaces, then there is an ordered basis β for V such that $[T]_\beta$ is a direct sum of k matrices.

Solution: True

See Theorem 5.25.

2. For each of the following linear operators T on the vector space V , determine whether the given subspace W is a T -invariant subspace of V .

- (c) $V = \mathbb{R}^3$, $T(a, b, c) = (a + b + c, a + b + c, a + b + c)$, and $W = \{(t, t, t) : t \in \mathbb{R}\}$

Solution: Yes

Note that $T((t, t, t)) = (3t, 3t, 3t) \in W$.

- (e) $V = M_{2 \times 2}(\mathbb{R})$, $T(A) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} A$, and $W = \{A \in V : A^t = A\}$

Solution: No

Note that $B = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \in W$, but $T(B) = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} \notin W$.

- 6.** For each linear operator T on the vector space V , find an ordered basis for the T -cyclic subspace generated by the vector z .

(a) $V = \mathbb{R}^4$, $T(a, b, c, d) = (a + b, b - c, a + c, a + d)$, and $z = e_1$.

Solution:

Note that $z = (1, 0, 0, 0)$, $T(z) = (1, 0, 1, 1)$, $T^2(z) = (1, -1, 2, 2)$, $T^3(z) = (0, -3, 3, 3)$.

Since $z, T(z), T^2(z)$ are linearly independent and $T^3(z) = -3T(z) + 3T^2(z)$, the desired ordered basis is $\{z, T(z), T^2(z)\}$.

(d) $V = M_{2 \times 2}(R)$, $T(A) = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} A$, and $z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Solution:

Note that $z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $T(z) = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$, $T^2(z) = \begin{pmatrix} 3 & 3 \\ 6 & 6 \end{pmatrix}$.

Since $z, T(z)$ are linearly independent and $T^2(z) = 3T(z)$, the desired ordered basis is $\{z, T(z)\}$.

- 9.** For each linear operator T and cyclic subspace W in Exercise 6, compute the characteristic polynomial of T_W in two ways, as in Example 6.

(a) $V = \mathbb{R}^4$, $T(a, b, c, d) = (a + b, b - c, a + c, a + d)$, and $z = e_1$.

Solution:

By means of Theorem 5.22, since $T^3(z) = -3T(z) + 3T^2(z)$, the characteristic polynomial of T_W is

$$(-1)^3(3t - 3t^2 + t^3).$$

By means of determinants, if $\beta = \{z, T(z), T^2(z)\}$, the characteristic polynomial of T_W is

$$\det([T_W]_{\beta} - tI) = \det \begin{pmatrix} -t & 0 & 0 \\ 1 & -t & -3 \\ 0 & 1 & 3-t \end{pmatrix} = -t^3 + 3t^2 - 3t.$$

(d) $V = M_{2 \times 2}(R)$, $T(A) = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} A$, and $z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Solution:

By means of Theorem 5.22, since $T^2(z) = 3T(z)$, the characteristic polynomial of T_W is

$$f(t) = (-1)^2(-3t + t^2).$$

By means of determinants, if $\beta = \{z, T(z)\}$, the characteristic polynomial of T_W is

$$\det([T_W]_{\beta} - tI) = \det \begin{pmatrix} -t & 0 \\ 1 & 3-t \end{pmatrix} = t^2 - 3t.$$

10. For each linear operator in Exercise 6, find the characteristic polynomial $f(t)$ of T , and verify that the characteristic polynomial of T_W (computed in Exercise 9) divides $f(t)$.

(a) $V = R^4$, $T(a, b, c, d) = (a + b, b - c, a + c, a + d)$, and $z = e_1$.

Solution:

Given the standard basis γ of V , the characteristic polynomial of T is computed as follows:

$$f(t) = \det([T]_{\gamma} - tI) = \det \begin{pmatrix} 1-t & 1 & 0 & 0 \\ 0 & 1-t & -1 & 0 \\ 1 & 0 & 1-t & 0 \\ 1 & 0 & 0 & 1-t \end{pmatrix} = (t-1)(t^3 - 3t^2 + 3t).$$

Obviously, the characteristic polynomial of T_W in Exercise 9 divides $f(t)$.

(d) $V = M_{2 \times 2}(R)$, $T(A) = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} A$, and $z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Solution:

Given the standard basis γ of V , the characteristic polynomial of T is computed as follows:

$$f(t) = \det([T]_{\gamma} - tI) = \det \begin{pmatrix} 1-t & 0 & 1 & 0 \\ 0 & 1-t & 0 & 1 \\ 2 & 0 & 2-t & 0 \\ 0 & 2 & 0 & 2-t \end{pmatrix} = t^2(t-3)^2.$$

Obviously, the characteristic polynomial of T_W in Exercise 9 divides $f(t)$.

17. Let A be an $n \times n$ matrix. Prove that

$$\dim(\text{span}(\{I_n, A, A^2, \dots\})) \leq n.$$

Solution:

By Cayley-Hamilton Theorem, there is the characteristic polynomial $f(t)$ of A such that

$$f(A) = a_0 I + a_1 A + \dots + a_n A^n = 0.$$

This shows that the given spanning set has dimension not bigger than n .

18. Let A be an $n \times n$ matrix with characteristic polynomial

$$f(t) = (-1)^n t^n + a_{n-1} t^{n-1} + \dots + a_1 t + a_0.$$

(a) Prove that A is invertible if and only if $a_0 \neq 0$.

Solution:

By Exercise 8 of Section 5.1, A is invertible if and only if 0 is not an eigenvalue of A . And 0 is not an eigenvalue of A if and only if the characteristic polynomial has a nonzero constant term.

(b) Prove that if A is invertible, then

$$A^{-1} = (-1/a_0)[(-1)^n A^{n-1} + a_{n-1} A^{n-2} + \dots + a_1 I_n].$$

Solution:

By Cayley-Hamilton, we have

$$(-1)^n A^n + a_{n-1} A^{n-1} + \dots + a_1 A + a_0 I = 0$$

and

$$(-1)^n A^n + a_{n-1} A^{n-1} + \dots + a_1 A = -a_0 I.$$

Since $a_0 \neq 0$ by (a), we can divide both sides by $-a_0$.

(c) Use (b) to compute A^{-1} for

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & -1 \end{pmatrix}.$$

Solution:

First we can find the characteristic polynomial of A as follows:

$$f(t) = (1-t)(2-t)(-1-t) = -t^3 + 2t^2 + t - 2.$$

Then by (b), we have

$$A^{-1} = -\frac{1}{2}A^2 + A + \frac{1}{2}I = \begin{pmatrix} 1 & -1 & -2 \\ 0 & \frac{1}{2} & \frac{3}{2} \\ 0 & 0 & -1 \end{pmatrix}.$$